

i want to add an insulin dosage calculator to my app.
 what factors do i need to consider and what details do
 they need to inout (e.g. current glucos
 insulin sensitivity, carbs, excercise adju
 adjustment etc.)

Upgrade

...

Yes — but because this is a **Type 1 diabetes app intended for children, adolescents and young adults**, I would treat an insulin calculator as a **high-risk clinical feature**, rather than simply adding a mathematical calculator. The calculator needs to use the person's **individual insulin prescription/settings**, with clear safety checks and clinician-directed parameters.

Current ADA guidance specifically describes mealtime insulin adjustment around **carbohydrate intake, current glucose, glucose trends, physical activity and sick-day management**, while ISPAD describes the core use of an insulin-to-carbohydrate ratio (ICR) and insulin sensitivity/correction factor (ISF).  Diabetes Journals +1

1. The basic calculator

For a standard meal bolus, your app could conceptually calculate:

Meal insulin + correction insulin = suggested bolus

A. Meal/carbohydrate dose

The user enters:

- **Carbohydrate amount (g)**
- **Insulin-to-carbohydrate ratio (ICR)**

For example:

$$60 \text{ g carbohydrates} \div 10 \text{ g/unit} = 6 \text{ units}$$

The ICR should be **user-specific and preferably entered as prescribed by their diabetes team**, rather than the app trying to determine it automatically.

ISPAD describes ICR as the number of grams of carbohydrate covered by 1 unit of insulin.  ISPAD

B. Correction dose

The user enters:

- **Current blood glucose**
- **Target blood glucose**
- **Insulin sensitivity factor (ISF) / correction factor**

Conceptually:

$$(\text{Current glucose} - \text{Target glucose}) \div \text{ISF} = \text{correction dose}$$

For example, if someone has a glucose of 12 mmol/L, a target of 6 mmol/L and an ISF of 3 mmol/L/unit:

$$(12 - 6) \div 3 = 2 \text{ units}$$

ISPAD gives the same basic correction calculation and notes that ISF can be based on an individual's total daily insulin dose, but for your app I would **not have the app independently establish the patient's prescribed ISF.**  ISPAD

2. What I would have the user enter

I'd divide the calculator into "**Personal settings**" and "**Today's calculation.**"

PERSONAL SETTINGS — entered once

These would be saved in the user's profile:

Setting	Why it matters
Insulin type	Rapid-acting vs other insulin
Insulin delivery method	Pen/syringe/pump
Insulin-to-carb ratio (ICR)	Calculates meal insulin
Insulin sensitivity factor (ISF)	Calculates correction insulin
Target glucose	Determines correction
Maximum single bolus	Safety limit
Insulin dose increment	e.g. 0.5-unit or 1-unit increments
Basal insulin information	Important context, although the calculator shouldn't automatically change basal insulin

Setting	Why it matters
Insulin duration/action time	Important if accounting for insulin-on-board
Low-glucose threshold	Allows the app to stop/restrict bolus recommendations
Ketone-management instructions	Safety/sick-day pathway

For children/adolescents, I would make it especially clear that these settings should come from the **parent/caregiver and/or diabetes healthcare team**, rather than being calculated by the app.

3. TODAY'S CALCULATION

I'd make the actual calculator screen very simple.

Step 1 — Current glucose

What is your current glucose?

- mmol/L
- Option to enter manually
- Potential future CGM integration

Also potentially:

Glucose trend:

↑ Rising / → Stable / ↓ Falling

This becomes particularly useful if you eventually integrate CGM data. ADA guidance recognises glucose trends as relevant to insulin adjustment.  Diabetes Journals

Step 2 — Carbohydrates

How many grams of carbohydrate are you eating?

_____ g

Then:

Meal insulin = carbohydrate ÷ ICR

Step 3 — Correction

The app uses:

- Current glucose
- Target glucose
- ISF

and calculates the correction component.

If glucose is **below target**, the app should **not simply produce a negative insulin dose**. It should move into a safety pathway instead.

4. VERY IMPORTANT: Insulin on board

I would strongly consider including this.

If someone took rapid-acting insulin recently, giving another full correction could result in **insulin stacking and hypoglycaemia**.

So potentially:

Have you taken rapid-acting insulin recently?

- No
- Yes → When?
- Amount taken: ____ units

Then:

Insulin on board (IOB): ____ units

A more sophisticated bolus calculator can account for active insulin rather than treating the person as if they have no previous insulin working.

This is one of the areas where I'd recommend getting your calculation logic reviewed by a **diabetes clinician/endocrinologist/pharmacist experienced in T1DM technology** before implementing it.

5. Exercise adjustment

I wouldn't make this a simple "exercise = -20%" button.

Exercise responses vary substantially according to the **type, intensity, duration and timing of exercise**, as well as insulin already active. ADA specifically notes that

exercise-related insulin adjustments need to be individualised.  Diabetes Journals

Instead, ask:

Are you planning to exercise soon?

- No
- Yes

If yes:

Exercise type

- Light
- Moderate
- Vigorous/high intensity

Expected duration

- <30 min
- 30–60 min
- 60 min

When will you exercise?

- Before eating
- Soon after eating
- Later

Current glucose

- Already captured

Glucose trend

- ↑
- →
- ↓

The app could then provide **safety guidance based on the user's clinician-defined plan**, rather than automatically inventing an insulin reduction.

This is particularly important because aerobic exercise can lower glucose, while high-intensity exercise can sometimes produce a temporary glucose rise.  Diabetes Journals

6. Illness/sick-day adjustment

I would definitely include a "I'm sick" option, but I would be cautious about making the app automatically calculate extra insulin.

Ask:

Are you currently unwell?

- No
- Yes

If yes:

Symptoms

- Fever
- Vomiting
- Diarrhoea
- Unable to eat/drink normally
- Other

Then:

Check ketones

- No ketone result
- Blood ketones: ____ mmol/L
- Urine ketones: ____

And the app should direct the user to an **appropriate sick-day protocol**, including when to check glucose/ketones, hydration, insulin guidance and when to seek urgent medical care.

This is particularly important because **T1DM + illness + insufficient insulin can progress to diabetic ketoacidosis (DKA)**.

ISPAD's sick-day guidance uses glucose and ketone levels to determine different management pathways, and specifically emphasises additional insulin/correction and reassessment in certain situations.  ISPAD

I would therefore make this a **sick-day guidance module**, rather than just:

"Sick = add 10% insulin."

7. Ketones should be a safety check

I'd actually put this **before the final dose is displayed**.

For example:

Safety check

Current glucose: ____ mmol/L

Ketones:

- Not checked
- <0.6 mmol/L
- 0.6–1.5 mmol/L
- 1.5 mmol/L

Depending on the result, the app can direct the user to the appropriate **clinician-approved sick-day guidance**.

ISPAD specifically incorporates ketone levels into exercise/sick-day decision-making and recommends avoiding exercise in certain hyperglycaemia/ketone situations.  ISPAD

8. Meals aren't only carbohydrates

This is something that could make your calculator more sophisticated.

ADA 2026 notes that insulin dosing may need to account for **fat and protein as well as carbohydrates**, although this is more complex and may be more feasible with insulin pumps than with multiple daily injections.  Diabetes Journals

For your **first prototype**, I would probably **not automatically calculate a fat/protein adjustment**.

Instead:

Carbohydrate: ____ g

Meal type: Regular / high-fat / high-protein

If high-fat/high-protein is selected, the app could provide an educational warning such as:

"High-fat or high-protein meals may affect glucose levels for several hours. Follow your usual diabetes-management plan."

That keeps the prototype safer and much easier to validate.

9. The final calculation screen

I would structure the result something like this:

Suggested mealtime dose

Carbohydrate dose:

5.0 units

Correction dose:

1.5 units

Insulin already active:

0.5 units

Exercise considerations:

Follow your personalised exercise plan

Total calculated dose:

6.0 units

Then:

Important: This is a calculation based on the insulin settings you entered. Check the result against your prescribed diabetes-management plan before administering insulin.

And have a prominent:

 **DO NOT ADMINISTER / CHECK GLUCOSE & FOLLOW HYPO GUIDANCE**

type pathway when glucose is low or other safety conditions are triggered.

10. Things I would NOT let the app decide automatically

This is probably the most important part for your MPharm project.

I would **not** have the prototype independently determine:

-  the user's ICR
-  the user's ISF
-  their basal insulin dose
-  changes to their long-acting insulin
-  automatic sick-day percentage increases
-  automatic exercise percentage reductions
-  correction doses when significant ketones are present without an appropriate safety

pathway

- ✗ insulin doses for someone who hasn't established their personalised insulin regimen
- ✗ a dose simply from current glucose

The ADA specifically emphasises that T1DM insulin treatment should incorporate individual factors including nutritional intake, glycemia, activity and sick-day management.  Diabetes Journals

11. I'd actually design your calculator in 3 levels

For **your particular dissertation prototype**, I think this would make the feature much more defensible:

LEVEL 1 — Basic meal calculator

Inputs:

- Current glucose
- Carbohydrates
- ICR
- Target glucose
- ISF

Output:

- Meal dose
- Correction dose
- Total calculated dose

LEVEL 2 — Safety adjustments

Additional information:

- Insulin on board
- Glucose trend
- Exercise planned?
- Recent exercise
- Illness
- Ketones

These don't necessarily automatically change the dose; they can **trigger appropriate warnings/guidance**.

LEVEL 3 — Emergency safety

The calculator should stop normal dose calculation and provide appropriate guidance if there are things such as:

- Hypoglycaemia
- Significant ketones
- Severe hyperglycaemia
- Vomiting/inability to keep fluids down
- Symptoms suggestive of DKA
- Conflicting/insufficient information

That fits very well with the **"Emergency Guidance"** feature you've already planned for your app.

A possible input flow for your Bubble prototype

I'd make it:

1. Glucose

Current glucose: ____ mmol/L

2. Meal

Carbohydrates: ____ g

3. Activity

Exercising soon? Yes / No

If yes:

Type: Light / Moderate / Vigorous

Duration: ____ min

4. Health

Are you currently sick? Yes / No

If yes:

Ketones: ____ mmol/L

5. Previous insulin

Rapid-acting insulin taken recently? Yes / No

If yes:

Dose: ____ units

Time: ____

6. Personal settings

ICR: ____ g/unit

ISF: ____ mmol/L/unit

Target glucose: ____ mmol/L

7. CHECK

Then show:

Calculated meal dose: X units

Correction: X units

Insulin on board: X units

Calculated total: X units

followed by the relevant safety message.

One important change I'd make to your thesis

Because your current project is a **prototype usability study rather than a clinical trial**, I would describe this feature as a "**personalised insulin bolus calculation aid**" rather than claiming that the app "determines the correct insulin dose."

Your methodology could explain that the calculator uses **user-entered, clinician-established parameters** (ICR, ISF and target glucose) to demonstrate the calculation process, while safety conditions direct users to predefined educational/emergency guidance.

That would also make the feature much easier to justify from a **user-centred design and patient-safety perspective**.

If you're adding this to the actual Bubble prototype, I can also ↪ map out the exact screens, input fields, buttons, formulas, warning messages and Bubble conditional logic for the calculator so you can build it directly.



ChatGPT can make mistakes. Check important info.